2 Million U.S. Wildfires:

Hotspot Identification and Fire-Spread Warning Model

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1 Introduction

Wildfires pose various health, environmental, and economic threats worldwide and are increasing in size and frequency. Beyond the instantaneous effects of the natural disaster, wildfire smoke carries a multitude of hazardous air pollutants that can spread further than its surrounding area [1]. Long-term wildfire health impacts show that exposure is not only linked to temporary issues like reduced lung function, but also to long-term increases in premature mortality and cardiovascular disease[6]. Moreover, wildfires can cause persistent mental health conditions, like depression, anxiety, and Post-Traumatic Stress Disorder (PTSD) [4,5,8]. On the economic side, insurers must maintain enough money reserved to cover potential claims caused by the wildfire damage. These reserve levels are much higher than lines with more stable and predictable losses, since wildfire damage cannot be easily predicted [13].

As the climate and global temperature continuously change in the future, understanding where wildfires occur and how they spread is a priority when discussing public health and safety. We build on this motivation by analyzing the open-source dataset WildfireDB, which links wildfire activity to various environmental characteristics across the United States. Our goal is to explore the patterns of wildfire occurrences and their spread. Using spatial hotspot analysis, we can identify wildfire regions where high-intensity wildfires tend to cluster and highlight potential areas that may face repeated exposure to wildfires and their long-term effects. Then, we want to predict whether a wildfire is likely to spread to its neighboring area, using environmental and weather predictors. These two analyses will provide insight into where intense wildfires are concentrated and which conditions contribute to wildfire spread.

2 Dataset Description

Our dataset contains over 17 million occurrences of U.S. wildfires from the years 2012 to 2018**,** capturing where fires occurred, how intense they were, and what the surrounding conditions looked like at the time. Each sample consists of a unique polygon identifier, the date and time of the fire, its neighboring polygon identifier, and the Fire Radiative Power (FRP) extracted from NASA's Visible Infrared Imaging Radiometer Suite (VIIRS), which collects global satellite observations across multiple visible and infrared wavelengths to monitor fire activity on Earth [10]. Each entry also represents a 375 x 375 meter cell detected by VIIRS. The cells are paired with a neighboring cell in the next time step to describe wildfire spread patterns.

Each fire also includes latitude and longitude values corresponding to the center of the VIIRS pixel to identify the exact physical location of each cell and allow us to map the wildfires. The

locational features are also used to describe the neighboring cell's coordinates.

In addition to fire intensity and location, the dataset includes a multitude of features describing their surrounding vegetation, topography information. The environmental layers come from sources such as LANDFIRE for vegetation and fuel data and Meteostat for daily weather observations. There are pre-engineered features containing zonal statistics, meaning each variable has many versions like minimum, maximum, mean and median. For instance, vegetation metrics such as Canopy Base Height or Canopy Bulk Density are further broken down to these 4 statistics. We observe that the same pattern appears for elevation-related features and weather variables like temperature, precipitation, wind speed, wind direction, and atmospheric pressure. Moreover, we see the same computations performed on the neighboring polygon, which assists us when comparing conditions around a spreading fire.

Altogether, the dataset accumulates to over 140 columns, combining identifying information, fire intensity, and a wide range of environmental descriptors like FRP, vegetation structure, fuel properties, elevation and slope statistics for both the active and neighboring cell [11,12].

3.1 Methods: Preprocessing

We begin by removing columns related to data collection, such as the polygon's ID, acquisition time, and shape, as well as similar columns describing the neighbor polygon. We also remove the average temperature column, which contains a large number of uninterpretable entries. We are able to do this without significantly impacting our models, as each row still contains information about the minimum and maximum temperatures recorded in the polygon. Finally, we impute all missing values with their column means.

3.2 Methods: Hotspot Analysis

In order to identify potential wildfire hotspots in the United States, we employ the K-means clustering method. We select the coordinates of the polygon and the fire intensity so that our clusters account for location as well as potential danger caused by more intense fires. In this way, our K-means implementation is a weighted modification.

In order to determine the optimal number of clusters, we select a subset of the data and perform the K-means clustering algorithm for K values ranging from 2 to 40. Once we have selected K values that most meaningfully cluster our data, we perform a full K-means clustering on our dataset using these values.

3.3 Methods: Spread Analysis

In order to predict which wildfires are likely to spread to neighboring regions, we employ the decision tree algorithm for binary classification. We use the column representing the FRP of the neighboring cell as our label, as this column indicates whether this wildfire has spread. We change any null values in this column to zeros, and convert any other numerical value to one.

Each row contains a combination of geographic, geological, and weather data, and decision trees are able to handle complex, non-linear relationships between various features [2]. For these reasons, we take all columns in the dataset other than our label as features. We need not scale any of these features, which contain a large range of values, as decision trees exploit splits in the data and these splits would be identical after scaling.

Because approximately 90% of the fires in our dataset did not spread, we are presented with a severe class imbalance. To address this, we add a weight column representing the inverse class representation and use this column to weigh our decisions in favor of the minority class. This weighted decision tree is rewarded for producing balanced predictions, increasing the utility of the model.

Before performing the decision tree algorithm on the full data, we select a subset of the data in order to determine the ideal maximum decision tree depth. We perform the following five times for each possible maximum depth:

1. We randomly select 70% of the data as training data, and designate 30% as testing data.
2. Using the training data, we construct a decision tree with a depth no larger than the specified maximum.
3. We evaluate and record the accuracy of the model on our testing data.
4. We compute the average accuracy for each maximum depth, and note the maximum depth that produced the highest average accuracy.

Finally, with our parameters determined, we split our full data into 70% training and 30% testing groups before generating a complete decision tree using the training data. We generate two trees: one weighted by inverse class representation and one completely unweighted.

4.1 Results: Hotspot Analysis

We found that the number of clusters with the highest silhouette score was two. When partitioned into two clusters, the wildfires appear to be split into east/west groups. Because the coordinates given for each wildfire are only given through a projection system, and this projection system is not shared, there is no way to map each fire, cluster, or cluster mean to an actual geographic location. However, the displayed collection of projected coordinates appear in the shape of the horizontal reflection of the contiguous United States, which we know to be the region represented by the data. This allows us to make conclusions about the general geography of our results. The divide between the two clusters is a low-fire zone representing the region just east of the Rocky Mountains. The eastern cluster covers one of the most tree-dense regions of the United States, while the area represented by the central divide between clusters covers a relatively tree-less region of the country [14].

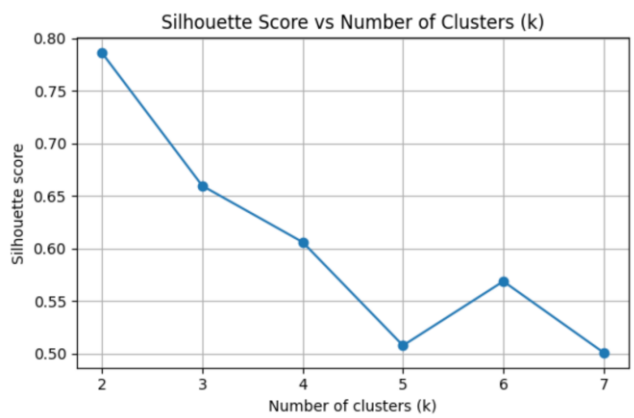

Figure 1: We find that k=2 gives the highest silhouette score.

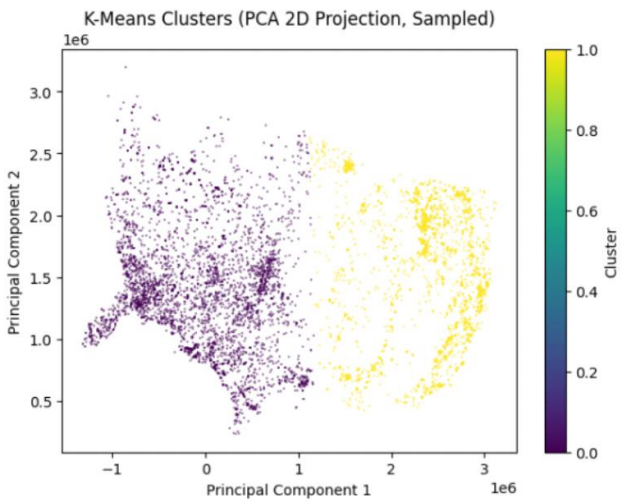

Figure 2: Results of K-means clustering with k=2

While the number of clusters that yielded the highest silhouette score was two, we found that two clusters represented regions too large to be considered wildfire hotspots. We extended our analysis to higher cluster numbers, with an overall range of 2 to 40, and found that 25 clusters also yielded a high silhouette score. These clusters, particularly the densest among them, more significantly represent wildfire hotpots. We determine that two especially dense clusters map to the Ozark Mountains highland region of Missouri, Arkansas, and Oklahoma, and a particularly dry grassland region in North Dakota [3,15].

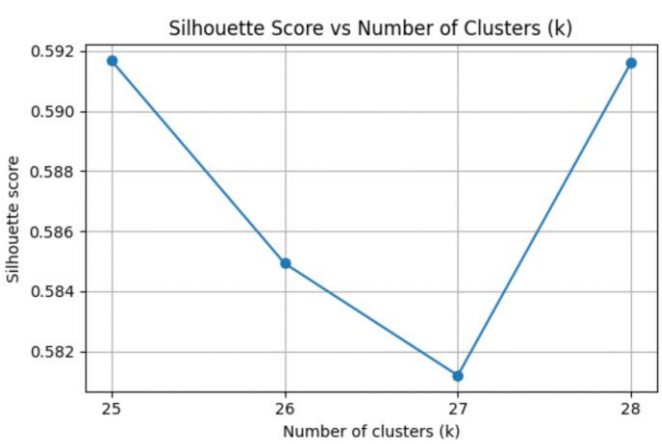

Figure 3: We find that k=25 gives a high silhouette score.

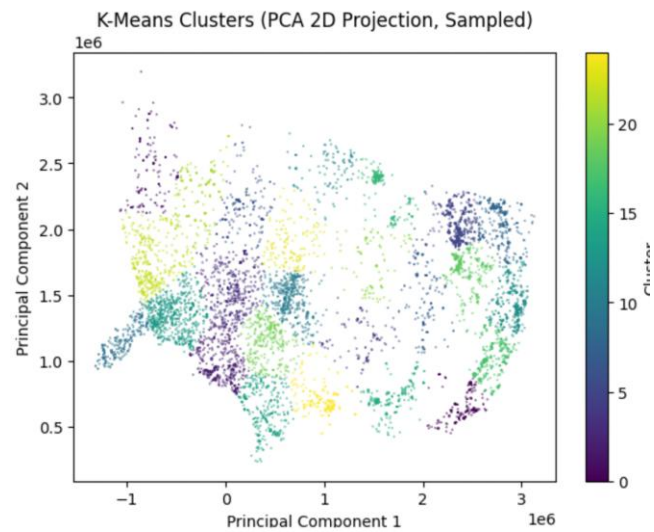

Figure 4: Results of K-Means clustering with k=25.

4.2 Results: Wildfire Spread Analysis

The decision tree generated with no weight parameter had depth zero and achieved 92.2% accuracy on the testing data. There were no decision nodes in the tree, which automatically guessed that a fire did not spread. This is largely due to the extreme class imbalance between fires that did and did not spread. Because the model could not produce a decision protocol that was more accurate than the representation percentage of the majority class, it did not add any decision nodes. This decision protocol produced zero true positives, that is, it never correctly predicted that a fire would spread. We take the harmonic mean of the decision tree's precision and recall to calculate an F1 score of 0, designating the model meaningless in real-world scenarios.

The decision tree weighted by inverse representation was of depth six, which was experimentally determined to be the optimal bound for the depth of the tree. This model achieved 72.3% accuracy on the testing data, which, although lower than the depth-zero model,

represents a much more balanced prediction profile. This is reflected in its F1 score of 0.78.

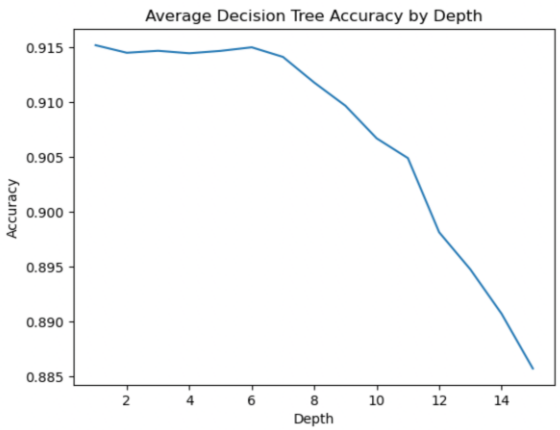

Figure 5: Results of the experiment to determine the ideal decision tree depth.

A large number of the dataset's 140 columns are utilized in this depth-six tree. Columns that appear repeatedly in the tree are latitude, longitude, and temperature, as well as the neighboring cell's canopy height, elevation, and existing vegetation cover. This indicates that relationships between a variety of features spanning the areas of geography, ecology, and climate provide predictive value to the model.

5.1 Conclusion: Hotspot Analysis

We conclude that there are concentrated occurrences of wildfires in the Ozark Mountains region and in a specific region of North Dakota. Our silhouette score of 0.59 indicates adequate clustering, but we believe it could be improved by adding more clusters. This study was limited by computational availability, so we would recommend extending the number of clusters in the K-means clustering algorithm in order to identify even smaller regions of concern for wildfires. This study was also limited in its ability to map fires to their exact locations, so in the future we would recommend identifying the exact locations of the regions specified here.

5.2 Conclusion: Wildfire Spread Analysis

We conclude that our decision tree model for predicting the spread of wildfires shows promise. Our F1 score of 0.78 outperforms a similar decision tree model trained for wildfire prediction which achieved an F1 score of 0.68 [9]. We very nearly match the performance of a far more complicated CNN-based model to predict wildfire spread that achieved an F1 score of 0.79 [7]. We believe that the decision tree binary classification algorithm was well-suited to our problem, as our data represented various fields with many possible complex relationships between them and our model exploited many of these relationships by including a varied selection of columns. This experiment, as with the hotspot analysis, was limited by computational availability. We believe that further research could improve accuracy by utilizing other, more computationally expensive forms of analysis for complex nonlinear relationships, such as multilayer perceptrons, to perform binary classification.

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